

# 3.0 - Info needed to make a relational model for a construction company

Sunday, October 26, 2025 6:51 PM

entities:

- Client details: Customer details(B2C company) / Business details(B2B company)
- Inventory details
- Employee info
- Project details
- Supplier details
- Finance info

relationships:

- Company client relationship (one client multiple projects)
- Relationship with one/many suppliers
- Contract details with each client
- Employee Project relationship
- Bills associated with each project

## 1. Client

- ClientID (*Primary Key*)
- ClientType (*B2C / B2B*)
- ClientName
- ContactPerson
- Phone
- Email
- Address
- CompanyName (*for B2B clients*)
- GSTNumber (*for B2B clients*)

## 2. Project

- ProjectID (*Primary Key*)
- ProjectName
- ClientID (*Foreign Key → Client*)
- StartDate
- EndDate
- Status (*Planned / Ongoing / Completed*)
- Location
- Budget

## 3. Employee

- EmployeeID (*Primary Key*)
- Name
- Role / Designation
- Phone
- Email
- HireDate
- Salary

## 4. Supplier

- SupplierID (*Primary Key*)
- SupplierName
- ContactPerson
- Phone
- Email
- Address

## 5. Inventory

- InventoryID (*Primary Key*)
- ItemName
- QuantityAvailable
- UnitPrice
- SupplierID (*Foreign Key → Supplier*)

## 6. Contract

- ContractID (*Primary Key*)
- ClientID (*Foreign Key → Client*)
- ProjectID (*Foreign Key → Project*)
- ContractDate
- TotalValue
- TermsAndConditions

## 7. Finance

- FinanceID (*Primary Key*)
- ProjectID (*Foreign Key → Project*)
- TotalExpense
- PaymentReceived
- OutstandingBalance

## 8. Bill

- BillID (*Primary Key*)
- ProjectID (*Foreign Key → Project*)
- BillDate
- Amount
- PaymentStatus (*Paid / Pending*)

## KEY RELATIONSHIPS

- **Client ↔ Project:**  
One client can have multiple projects (1-to-Many).
- **Project ↔ Employee:**  
Many employees can work on many projects (Many-to-Many).
- **Project ↔ Contract:**  
Each project has one contract with a client (1-to-1 or 1-to-Many depending on scenario).
- **Supplier ↔ Inventory:**  
One supplier can supply many inventory items (1-to-Many).
- **Project ↔ Finance:**  
One project has one finance record (1-to-1).
- **Project ↔ Bill:**  
One project can have multiple bills (1-to-Many).